0. Axiomas

We impose the following axioms on the choice correspondence c.

Axiom 1 (Pseudo-WARP). For all $A, B \in \mathcal{A}$, if $f \in c(B) \cap A$ and $c(A) \subseteq B$, then $f \in c(A)$.

Axiom 2 (Independence). For every $A \in \mathcal{A}$, $\lambda \in (0,1)$, and $h \in \mathcal{F}$,[1] $c\big(\lambda A + (1-\lambda)h\big) = \lambda c(A) + (1-\lambda)h$.

Axiom 3 (Continuity). The set $\big\{(A,f) \in \mathcal{A} \times \mathcal{F} : f \in c(A)\big\}$ is closed.

Axiom 4 (Indirect Ambiguity Aversion). For every $A \in \mathcal{A}$, $\lambda \in [0,1]$, and $f, g, h \in \mathcal{F}$, if $f \notin c\big(A \cup \{h\}\big)$ and $f \notin c\big(A \cup \{g\}\big)$, then $f \notin c\big(A \cup \{\lambda g + (1-\lambda)h\}\big)$.

Axiom 5 (No-Hedging). For every $A \in \mathcal{A}$, $\lambda \in [0,1]$, and $g, h \in \mathcal{F}$, if $\{g,h\} \subseteq A$, then $c(A) \subseteq c\big(A \cup \{\lambda g + (1-\lambda)h\}\big)$.

Axiom 6 (Best and Worst). For every $A \in \mathcal{A}$,

$$\{x^*\} = c\big(A \cup \{x^*\}\big) \quad \text{and} \quad c(A) = c\big(A \cup \{x_*\}\big).$$

Axiom 7 (One-Side Monotonicity). For every $A \in \mathcal{A}$ and every $f, g \in \mathcal{F}$, if $f \notin c\big(A \cup \{g(s)\}\big)$ for every $s \in S$, then $f \notin c\big(A \cup \{g\}\big)$.

1. Theorems

Theorem 1.1 (Multi-Prior Multi-Utility in a Finite Prize Space) *The following statements are equivalent:*

(1) c has a pseudo-rational multi-taste multi-prior representation.

(2) c satisfies Pseudo-WARP, Independence, Continuity, and Indirect Ambiguity Aversion, as well as No-Hedging, Best and Worst, and One-Side Monotonicity.

Proof.

Necessity. For brevity, we prove only the necessity of Indirect Ambiguity Aversion and No-Hedging; the necessity of the remaining axioms is standard.

Sufficiency. Fix $f \in \mathcal{F}$, and let $A_f \in \mathcal{A}$ be maximal under set inclusion among the menus A such that $f \in c(A)$. Define the binary relation $\succsim_{A_f}$ on $\mathcal{F}$ by

$$g \succsim_{A_f} h \iff \forall \alpha \in [0,1],\ \forall x \in \{x_*, x^*\},\ \alpha g + (1-\alpha)x \in A_f \implies \alpha h + (1-\alpha)x \in A_f.$$

It is immediate that this relation satisfies transitiviy and that, by No-Hedging, A_f is convex.

[1] We abuse notation and write $\lambda A + (1-\lambda)h$ for the set $\{\lambda f + (1-\lambda)h : f \in A\}$.

Lemma 1 *The relation $\succsim_{A_f}$ satisfies completeness.*

Proof. For every act $j \in \mathcal{F}$, define the sets

$$\Gamma_*(j) := \{\beta \in [0,1] : \beta j + (1-\beta)x_* \in A_f\}$$

and

$$\Gamma^*(j) := \{\beta \in [0,1] : \beta j + (1-\beta)x^* \in A_f\}.$$

It is immediate that

$$g \succsim_{A_f} h \iff \Gamma_*(g) \subseteq \Gamma_*(h) \quad \text{and} \quad \Gamma^*(g) \subseteq \Gamma^*(h).$$

We will show that, in every possible case, either

$$\Gamma_*(g) \subseteq \Gamma_*(h) \quad \text{and} \quad \Gamma^*(g) \subseteq \Gamma^*(h),$$

or

$$\Gamma_*(h) \subseteq \Gamma_*(g) \quad \text{and} \quad \Gamma^*(h) \subseteq \Gamma^*(g),$$

which is equivalent to saying that the relation is complete.

First, consider some necessary claims. For the following argument, assume $f \notin \{x_*, x^*\}$, identifying the outcomes with their constant acts.

Claim 1 *$x_* \in A_f$ and $x^* \notin A_f$.*

Proof. Since $f \in c(A_f)$, Best and Worst gives $c(\{f, x_*\}) = \{f\}$. If $x_* \notin A_f$, then $x_* \notin c(A_f \cup \{x_*\})$: otherwise, Pseudo-WARP applied to the smaller menu $\{f, x_*\}$ would imply $x_* \in c(\{f, x_*\})$, a contradiction. Therefore $c(A_f \cup \{x_*\}) \subseteq A_f$. Since $f \in c(A_f) \cap (A_f \cup \{x_*\})$, Pseudo-WARP now implies $f \in c(A_f \cup \{x_*\})$. By maximality of A_f, this implies $x_* \in A_f$, a contradiction. Thus $x_* \in A_f$.

For the best case, suppose, toward a contradiction, that $x^* \in A_f$. Then $f \in c(A_f) \cap \{f, x^*\}$ and, by the hypothesis, $c(\{f, x^*\}) \subseteq A_f$. Pseudo-WARP implies $f \in c(\{f, x^*\})$, contradicting Best and Worst, which gives $c(\{f, x^*\}) = \{x^*\}$. ■

Claim 2 *If $g \notin A_f$ and $h \notin A_f$, then $\alpha g + (1-\alpha)h \notin A_f$ for every $\alpha \in (0,1)$.*

Proof. Suppose $g \notin A_f$ and $h \notin A_f$. By maximality of A_f, $f \notin c(A_f \cup \{g\})$ and $f \notin c(A_f \cup \{h\})$. Therefore

$$f \notin c(A_f \cup \{g\}) \cup c(A_f \cup \{h\}).$$

Then Indirect Ambiguity Aversion implies that

$$f \notin c(A_f \cup \{\alpha g + (1-\alpha)h\}).$$

Since $f \in c(A_f)$, it follows that $\alpha g + (1-\alpha)h \notin A_f$. ■

Claim 3 *For every $j \in \mathcal{F}$, the sets $\Gamma_*(j)$ and $\Gamma^*(j)$ are closed and convex.*

Proof. By Continuity, A_f is closed. Since mixtures depend continuously on their weights, it is easy to see that $\Gamma_*(j)$ and $\Gamma^*(j)$ are both closed.

For the convexity argument, consider $\alpha_1, \alpha_2 \in \Gamma_*(j)$ and fix $\lambda \in [0,1]$. Showing that $\gamma := \lambda\alpha_1 + (1-\lambda)\alpha_2 \in \Gamma_*(j)$ is equivalent to showing that $\gamma j + (1-\gamma)x_* \in A_f$. This follows immediately from the convexity of A_f, since

$$\gamma j + (1-\gamma)x_* = \lambda\big(\alpha_1 j + (1-\alpha_1)x_*\big) + (1-\lambda)\big(\alpha_2 j + (1-\alpha_2)x_*\big).$$

The same argument, with x^* replacing x_*, shows that $\Gamma^*(j)$ is convex. ■

Fix $g, h \in \mathcal{F}$. Notice that, if $g \in A_f$, Claim 3 implies that $\Gamma_*(g)$ and $\Gamma^*(g)$ are closed intervals. Since $0, 1 \in \Gamma_*(g)$, we have $\Gamma_*(g) = [0,1]$. Also, $1 \in \Gamma^*(g)$ and $0 \notin \Gamma^*(g)$ by Claim 1. Thus there exists $\alpha^* \in (0,1]$ such that $\Gamma^*(g) = [\alpha^*, 1]$.

Conversely, if $g \notin A_f$, then $0, 1 \notin \Gamma^*(g)$. By Claims 1 and 2, for every $\lambda \in (0,1)$,

$$\lambda g + (1-\lambda)x^* \notin A_f,$$

so $\Gamma^*(g) = \emptyset$. By the same token, Claims 1 and 3, together with $0 \in \Gamma_*(g)$ and $1 \notin \Gamma_*(g)$, imply that there exists $\alpha_* \in [0,1)$ such that $\Gamma_*(g) = [0, \alpha_*]$. Given these possibilities, we can determine the sets for both g and h in every possible case.

Case 1: $g, h \in A_f$**.** In this case,

$$\begin{aligned} \Gamma_*(g) &= [0,1], \quad & \Gamma_*(h) &= [0,1], \\ \Gamma^*(g) &= [\alpha^*, 1], \quad & \Gamma^*(h) &= [\beta^*, 1]. \end{aligned}$$

If $\alpha^* \geq \beta^*$, then $\Gamma^*(g) \subseteq \Gamma^*(h)$, so $g \succsim_{A_f} h$. If $\alpha^* \leq \beta^*$, the reverse inclusions hold, so $h \succsim_{A_f} g$.

Case 2: $g, h \notin A_f$**.** In this case,

$$\begin{aligned} \Gamma_*(g) &= [0, \alpha_*], \quad & \Gamma_*(h) &= [0, \beta_*], \\ \Gamma^*(g) &= \emptyset, \quad & \Gamma^*(h) &= \emptyset. \end{aligned}$$

If $\alpha_* \geq \beta_*$, then $\Gamma_*(h) \subseteq \Gamma_*(g)$, so $h \succsim_{A_f} g$. If $\alpha_* \leq \beta_*$, the reverse inclusions hold, so $g \succsim_{A_f} h$.

Case 3: $g \in A_f$ **and** $h \notin A_f$**.** In this case,

$$\begin{aligned} \Gamma_*(g) &= [0,1], \quad & \Gamma_*(h) &= [0, \beta_*], \\ \Gamma^*(g) &= [\alpha^*, 1], \quad & \Gamma^*(h) &= \emptyset. \end{aligned}$$

Thus $\Gamma_*(h) \subseteq \Gamma_*(g)$ and $\Gamma^*(h) \subseteq \Gamma^*(g)$, so $h \succsim_{A_f} g$.

Case 4: $g \notin A_f$ **and** $h \in A_f$**.** In this case,

$$\begin{aligned} \Gamma_*(g) &= [0, \alpha_*], \quad & \Gamma_*(h) &= [0,1], \\ \Gamma^*(g) &= \emptyset, \quad & \Gamma^*(h) &= [\beta^*, 1]. \end{aligned}$$

Thus $\Gamma_*(g) \subseteq \Gamma_*(h)$ and $\Gamma^*(g) \subseteq \Gamma^*(h)$, so $g \succsim_{A_f} h$.

Therefore, in every case, either $g \succsim_{A_f} h$ or $h \succsim_{A_f} g$. Hence, the relation is complete. ■

Lemma 2 *The relation $\succsim_{A_f}$ is transitive.*

Proof. If $g \succsim_{A_f} h$ and $h \succsim_{A_f} k$, then each benchmark mixture of g that belongs to A_f implies that the corresponding mixture of h belongs to A_f, and hence that the corresponding mixture of k belongs to A_f. Thus $g \succsim_{A_f} k$. ■

Lemma 3 *The relation $\succsim_{A_f}$ satisfies continuity.*

Proof. Let (g^m) and (h^m) be two sequences in $\mathcal{F}$ such that $g^m \succsim_{A_f} h^m$ for every $m \in \mathbb{N}$, $g^m \to g$, and $h^m \to h$. We must show that $g \succsim_{A_f} h$. As in Lemma 1, assume $f \notin \{x_*, x^*\}$.

Suppose $\alpha \in (0,1)$ and $x \in \{x_*, x^*\}$ are such that

$$\alpha g + (1-\alpha)x \in A_f.$$

There are two possible cases: either $x = x_*$ or $x = x^*$. Consider the following necessary claims.

Claim 4 *Let $j \in \mathcal{F}$. If $j \notin A_f$, then $j \in c(A_f \cup \{j\})$.*

Proof. Suppose $j \notin A_f$ and $j \notin c(A_f \cup \{j\})$. Then $c(A_f \cup \{j\}) \subseteq A_f$ and $f \in c(A_f) \cap (A_f \cup \{j\})$. Pseudo-WARP implies that $f \in c(A_f \cup \{j\})$. By maximality of A_f, this implies $j \in A_f$, a contradiction. ■

Claim 5 *Let $j, k \in A_f$ be distinct acts. Then, for every $\gamma \in (0,1)$, $\gamma j + (1-\gamma)k \notin c(A_f)$.*

Proof. Suppose $j, k \in A_f$ and $j \neq k$. By convexity of A_f, $\gamma j + (1-\gamma)k \in A_f$. By Indirect Ambiguity Aversion, we obtain

$$\gamma j + (1-\gamma)k \notin c(\{j, k, \gamma j + (1-\gamma)k\}).$$

Suppose, toward a contradiction, that $\gamma j + (1-\gamma)k \in c(A_f)$. Then

$$\gamma j + (1-\gamma)k \in c(A_f) \cap \{j, k, \gamma j + (1-\gamma)k\},$$

and $c(\{j, k, \gamma j + (1-\gamma)k\}) \subseteq A_f$. Pseudo-WARP implies

$$\gamma j + (1-\gamma)k \in c(\{j, k, \gamma j + (1-\gamma)k\}),$$

a contradiction. ■

Claim 6 *If $\alpha g + (1-\alpha)x \in A_f$ and $g^m \to g$, then, for every $\lambda \in (0,1)$, there exists $M \in \mathbb{N}$ such that*

$$\lambda\big(\alpha g^m + (1-\alpha)x\big) + (1-\lambda)x_* \in A_f \quad \textit{for every } m \geq M.$$

This also holds when $\alpha = 1$.

Proof. Suppose, toward a contradiction, that there exists $\lambda \in (0,1)$ such that, for every $M \in \mathbb{N}$, there exists $m \geq M$ with

$$\lambda\big(\alpha g^m + (1-\alpha)x\big) + (1-\lambda)x_* \notin A_f.$$

Define the set

$$\mathbb{N}' := \big\{m \in \mathbb{N} : \lambda\big(\alpha g^m + (1-\alpha)x\big) + (1-\lambda)x_* \notin A_f\big\},$$

which is infinite by assumption. By Claim 4, for every $m \in \mathbb{N}'$,

$$\begin{aligned}&\lambda\big(\alpha g^m + (1-\alpha)x\big) + (1-\lambda)x_* \\ &\qquad \in c\Big(A_f \cup \big\{\lambda\big(\alpha g^m + (1-\alpha)x\big) + (1-\lambda)x_*\big\}\Big).\end{aligned}$$

Since $\alpha g + (1-\alpha)x \in A_f$, $x_* \in A_f$, and A_f is convex,

$$\lambda\big(\alpha g + (1-\alpha)x\big) + (1-\lambda)x_* \in A_f.$$

Moreover, $g^m \to g$, so the menus

$$A_f \cup \big\{\lambda\big(\alpha g^m + (1-\alpha)x\big) + (1-\lambda)x_*\big\}$$

converge to A_f in the Hausdorff metric. Passing to the subsequence indexed by $\mathbb{N}'$, Continuity gives

$$\lambda\big(\alpha g + (1-\alpha)x\big) + (1-\lambda)x_* \in c(A_f). \tag{$**$}$$

We proceed to show that this generates a contradiction.

First suppose $\alpha g + (1-\alpha)x \neq x_*$. By $(**)$,

$$\begin{aligned}&\lambda\big(\alpha g + (1-\alpha)x\big) + (1-\lambda)x_* \\ &\qquad \in c(A_f) \cap \big\{\alpha g + (1-\alpha)x,\ x_*,\ \lambda\big(\alpha g + (1-\alpha)x\big) + (1-\lambda)x_*\big\},\end{aligned}$$

and

$$c\Big(\big\{\alpha g + (1-\alpha)x,\ x_*,\ \lambda\big(\alpha g + (1-\alpha)x\big) + (1-\lambda)x_*\big\}\Big) \subseteq A_f.$$

Thus Pseudo-WARP implies that

$$\begin{aligned}&\lambda\big(\alpha g + (1-\alpha)x\big) + (1-\lambda)x_* \\ &\qquad \in c\Big(\big\{\alpha g + (1-\alpha)x,\ x_*,\ \lambda\big(\alpha g + (1-\alpha)x\big) + (1-\lambda)x_*\big\}\Big).\end{aligned}$$

However, that directly contradicts Claim 5.

If $\alpha g + (1-\alpha)x = x_*$, then $\lambda(\alpha g + (1-\alpha)x) + (1-\lambda)x_* = x_*$. Pseudo-WARP applied to $\{f, x_*\} \subseteq A_f$ and $(**)$ implies $x_* \in c(\{f, x_*\})$, contradicting Best and Worst and $f \neq x_*$. The same argument applies when $\alpha = 1$. ■

Case 1: $x = x_*$**.** Suppose $\alpha g + (1-\alpha)x_* \in A_f$. For every fixed $\lambda \in (0,1)$, Claim 6 gives $M \in \mathbb{N}$ such that, for every $m \geq M$,

$$\lambda\big(\alpha g^m + (1-\alpha)x_*\big) + (1-\lambda)x_* = \lambda\alpha g^m + (1-\lambda\alpha)x_* \in A_f.$$

Since $g^m \succsim_{A_f} h^m$ for every m, it follows that

$$\lambda\alpha h^m + (1-\lambda\alpha)x_* \in A_f \quad \text{for every } m \geq M.$$

By closedness of A_f, passing to the limit as $m \to \infty$ gives $\lambda\alpha h + (1-\lambda\alpha)x_* \in A_f$. Letting $\lambda \to 1$, another application of closedness gives

$$\alpha h + (1-\alpha)x_* \in A_f.$$

Case 2: $x = x^*$**.** Suppose $\alpha g + (1-\alpha)x^* \in A_f$. By Claims 1 and 2, $g \in A_f$. Define $\alpha_t := \alpha t + (1-t)$ for every $t \in (0,1)$, and observe that

$$\alpha_t g + (1-\alpha_t)x^* = t\big(\alpha g + (1-\alpha)x^*\big) + (1-t)g. \tag{$*$}$$

Consider now the following claims.

Claim 7 *For every $t \in (0,1)$, $\alpha_t g + (1-\alpha_t)x^* \in A_f$.*

Proof. This follows immediately from $g \in A_f$ and $\alpha g + (1-\alpha)x^* \in A_f$: since A_f is convex, $(*)$ gives the desired conclusion. ■

Claim 8 *If $g^m \to g$ and $\alpha g + (1-\alpha)x^* \in A_f$, then, for every fixed $t \in (0,1)$, there exists $M \in \mathbb{N}$ such that*

$$\alpha_t g^m + (1-\alpha_t)x^* \in A_f \quad \textit{for every } m \geq M.$$

Proof. The proof is similar to that of Claim 6. Fix $t \in (0,1)$ and suppose, toward a contradiction, that the statement does not hold. Then the set

$$\mathbb{N}' := \{m \in \mathbb{N} : \alpha_t g^m + (1-\alpha_t)x^* \notin A_f\}$$

is infinite. By Claim 4, for every $m \in \mathbb{N}'$,

$$\alpha_t g^m + (1-\alpha_t)x^* \in c\big(A_f \cup \{\alpha_t g^m + (1-\alpha_t)x^*\}\big).$$

By Claim 7, the limit $\alpha_t g + (1-\alpha_t)x^*$ belongs to A_f, so these menus converge to A_f. Passing to the appropriate subsequence, Continuity implies

$$\alpha_t g + (1-\alpha_t)x^* \in c(A_f).$$

But $g \in A_f$ and $x^* \notin A_f$ imply $g \neq x^*$; since $\alpha \in (0,1)$, the acts g and $\alpha g + (1-\alpha)x^*$ are therefore distinct. Equation $(*)$ and Claim 5 now give a contradiction. ■

For each fixed $t \in (0,1)$, Claim 8 and $g^m \succsim_{A_f} h^m$ imply that $\alpha_t h^m + (1-\alpha_t)x^* \in A_f$ for all sufficiently large m. By closedness of A_f,

$$\alpha_t h + (1-\alpha_t)x^* \in A_f.$$

Define $\Gamma^*(h)$ as in Lemma 1. Since α_t ranges over $(\alpha,1)$ as t ranges over $(0,1)$, we have $(\alpha,1) \subseteq \Gamma^*(h)$. Define

$$\phi : [0,1] \to \mathcal{F}, \qquad \phi(\beta) := \beta h + (1-\beta)x^*.$$

Then $\phi^{-1}(A_f) = \Gamma^*(h)$. Since A_f is closed and ϕ is continuous, its preimage is closed in $[0,1]$. Therefore $\alpha \in \Gamma^*(h)$, that is, $\alpha h + (1-\alpha)x^* \in A_f$.

Finally, if $\alpha = 0$, the required implication is immediate because both mixtures equal x. If $\alpha = 1$, the premise is $g \in A_f$; the argument of Case 1, using Claim 6 with $\alpha = 1$, gives $h \in A_f$. Thus all $\alpha \in [0,1]$ are covered, and $g \succsim_{A_f} h$.

We next record further properties of $\succsim_{A_f}$ that will be used to construct an affine representation.

Claim 9 *For every $g \in \mathcal{F}$, $x^* \succsim_{A_f} g \succsim_{A_f} x_*$.*

Proof. We first show that $x^* \succsim_{A_f} g$. Fix $\alpha \in [0,1]$ and $x \in \{x_*, x^*\}$, and suppose that

$$\alpha x^* + (1-\alpha)x \in A_f.$$

If $x = x^*$, the premise cannot hold because $x^* \notin A_f$. Thus it remains to consider $x = x_*$. If $g \in A_f$, convexity of A_f gives $\alpha g + (1-\alpha)x_* \in A_f$. Suppose instead that $g \notin A_f$. The cases $\alpha = 0$ and $\alpha = 1$ are immediate: the target is x_* in the former case, while the premise is false in the latter. For $\alpha \in (0,1)$, suppose, toward a contradiction, that $\alpha g + (1-\alpha)x_* \notin A_f$. By Claim 4,

$$\alpha g + (1-\alpha)x_* \in c\big(A_f \cup \{\alpha g + (1-\alpha)x_*\}\big).$$

Best and Worst gives $c(\{x^*, g\}) = \{x^*\}$, and Independence therefore yields

$$c\big(\{\alpha x^* + (1-\alpha)x_*,\ \alpha g + (1-\alpha)x_*\}\big) = \{\alpha x^* + (1-\alpha)x_*\}.$$

Since $\alpha x^* + (1-\alpha)x_* \in A_f$, Pseudo-WARP applied to $A_f \cup \{\alpha g + (1-\alpha)x_*\}$ and $\{\alpha x^* + (1-\alpha)x_*, \alpha g + (1-\alpha)x_*\}$ implies that $\alpha g + (1-\alpha)x_* \in c\big(\{\alpha x^* + (1-\alpha)x_*, \alpha g + (1-\alpha)x_*\}\big)$, a contradiction. Hence $\alpha g + (1-\alpha)x_* \in A_f$, proving $x^* \succsim_{A_f} g$.

We now show that $g \succsim_{A_f} x_*$. Fix $\alpha \in [0,1]$ and $x \in \{x_*, x^*\}$, and suppose that $\alpha g + (1-\alpha)x \in A_f$. If $x = x_*$, the desired conclusion follows from $x_* \in A_f$ (Claim 1). If $x = x^*$, the case $\alpha = 0$ cannot arise because $x^* \notin A_f$, and the case $\alpha = 1$ follows again from $x_* \in A_f$. Consider $\alpha \in (0,1)$. Claim 2, together with $x^* \notin A_f$, implies that $g \in A_f$. Suppose, toward a contradiction, that $\alpha x_* + (1-\alpha)x^* \notin A_f$. By Claim 4,

$$\alpha x_* + (1-\alpha)x^* \in c\big(A_f \cup \{\alpha x_* + (1-\alpha)x^*\}\big).$$

By Best and Worst, $c(\{g, x_*\}) = \{g\}$, and Independence implies

$$c\big(\{\alpha g + (1-\alpha)x^*, \alpha x_* + (1-\alpha)x^*\}\big) = \{\alpha g + (1-\alpha)x^*\}.$$

Since $\alpha g + (1-\alpha)x^* \in A_f$, Pseudo-WARP applied to $A_f \cup \{\alpha x_* + (1-\alpha)x^*\}$ and $\{\alpha g + (1-\alpha)x^*, \alpha x_* + (1-\alpha)x^*\}$ implies that $\alpha x_* + (1-\alpha)x^*$ is chosen from the latter menu, contradicting the preceding equality. Therefore $\alpha x_* + (1-\alpha)x^* \in A_f$, as required. This proves $g \succsim_{A_f} x_*$. ■

Claim 10 *For every $g \in \mathcal{F}$, there exists $\alpha_g \in [0,1]$ such that $g \sim_{A_f} \alpha_g x^* + (1-\alpha_g)x_*$.*

Proof. Fix any $g \in \mathcal{F}$. Consider the sets

$$\{\lambda \in [0,1] : q_\lambda \succsim_{A_f} g\} \quad \text{and} \quad \{\lambda \in [0,1] : g \succsim_{A_f} q_\lambda\}.$$

By completeness (Lemma 1), their union is $[0,1]$. By Claim 9, both sets are nonempty. The continuity of $\succsim_{A_f}$ established in Lemma 3 and the continuity of $\succsim_{A_f}$ imply that both sets are closed in $[0,1]$. Since $[0,1]$ is connected, the two sets cannot be disjoint; otherwise, they would form a nontrivial separation of $[0,1]$. Thus there exists α_g in their intersection, so that $q_{\alpha_g} \succsim_{A_f} g$ and $g \succsim_{A_f} q_{\alpha_g}$. Therefore $g \sim_{A_f} q_{\alpha_g}$. ■

Claim 11 *Define $q_\lambda := \lambda x^* + (1-\lambda)x_*$ for every $\lambda \in [0,1]$. Then $\alpha \geq \lambda$ if and only if $q_\alpha \succsim_{A_f} q_\lambda$.*

Proof. Since A_f is closed and convex, and $x_* \in A_f$ while $x^* \notin A_f$, its intersection with the segment $\{q_\lambda : \lambda \in [0,1]\}$ is of the form $\{q_\lambda : \lambda \in [0,\theta]\}$ for some $\theta \in [0,1)$. We first show that $\theta > 0$. Suppose instead that $q_\lambda \notin A_f$ for every $\lambda \in (0,1]$. By Claim 4,

$$q_\lambda \in c(A_f \cup \{q_\lambda\}) \quad \text{for every } \lambda \in (0,1].$$

As $\lambda \to 0$, these menus converge to A_f and $q_\lambda \to x_*$. Continuity would imply $x_* \in c(A_f)$. But Best and Worst gives $c(\{f, x_*\}) = \{f\}$, and Pseudo-WARP would then imply $x_* \in c(\{f, x_*\})$, a contradiction. Thus $\theta \in (0,1)$ and

$$q_\lambda \in A_f \quad \Longleftrightarrow \quad \lambda \leq \theta.$$

For every $\lambda, \beta \in [0,1]$,

$$\beta q_\lambda + (1-\beta)x^* = q_{1-\beta(1-\lambda)}, \qquad \beta q_\lambda + (1-\beta)x_* = q_{\beta\lambda}.$$

It follows that, if $\lambda \leq \theta$,

$$\Gamma^*(q_\lambda) = \left[\frac{1-\theta}{1-\lambda}, 1\right], \qquad \Gamma_*(q_\lambda) = [0,1],$$

whereas, if $\lambda > \theta$,

$$\Gamma^*(q_\lambda) = \emptyset, \qquad \Gamma_*(q_\lambda) = \left[0, \frac{\theta}{\lambda}\right].$$

Let $\alpha = \mu + \lambda$, where $\lambda > 0$ and $\alpha \leq 1$. If $\mu < \theta$, take λ small enough that $\alpha \leq \theta$. Then the formulas above imply $\Gamma^*(q_\alpha) \subsetneq \Gamma^*(q_\mu)$, while $\Gamma_*(q_\alpha) = \Gamma_*(q_\mu) = [0,1]$. If $\mu = \theta$, then $\Gamma^*(q_\alpha) = \emptyset \subsetneq \Gamma^*(q_\mu)$ and $\Gamma_*(q_\alpha) = [0, \theta/\alpha] \subsetneq [0,1] = \Gamma_*(q_\mu)$. Finally, if $\mu > \theta$, then $\Gamma^*(q_\alpha) = \Gamma^*(q_\mu) = \emptyset$ and $\Gamma_*(q_\alpha) = [0, \theta/\alpha] \subsetneq [0, \theta/\mu] = \Gamma_*(q_\mu)$. Thus, in each case, the characterization of $\succsim_{A_f}$ in Lemma 1 gives

$$q_{\mu+\lambda} \succsim_{A_f} q_\mu,$$

but not the reverse. Given any $\alpha > \mu$, choose a finite increasing sequence from μ to α whose increments satisfy the conditions above, using θ as an intermediate point if needed. Applying the preceding comparison at each increment and using transitivity gives $q_\alpha \succsim_{A_f} q_\mu$. The reverse cannot hold, since it would imply the reverse of one of the strict comparisons along the way. Therefore,

$$q_\alpha \succsim_{A_f} q_\mu \quad \Longleftrightarrow \quad \alpha \geq \mu,$$

■

Claim 12 *Let $g, h \in A_f$ and $\lambda \in [0,1]$. Then*

$$\frac{1}{\min \Gamma^*(\lambda g + (1-\lambda)h)} = \frac{\lambda}{\min \Gamma^*(g)} + \frac{1-\lambda}{\min \Gamma^*(h)}.$$

Proof. Fix $g, h \in A_f$ and $\lambda \in [0,1]$, and define

$$a := \min \Gamma^*(g), \qquad b := \min \Gamma^*(h).$$

Since $g, h \in A_f$ and $x^* \notin A_f$, we have $a, b \in (0,1]$, and, by construction,

$$ag + (1-a)x^* \in A_f, \qquad bh + (1-b)x^* \in A_f.$$

If $\lambda = 0$ or $\lambda = 1$, the result follows immediately. Thus, assume $\lambda \in (0,1)$, and define

$$t := \left(\frac{\lambda}{a} + \frac{1-\lambda}{b}\right)^{-1}, \qquad k := \lambda g + (1-\lambda)h, \qquad e_g := ag + (1-a)x^*, \qquad e_h := bh + (1-b)x^*.$$

We show first that $t \in \Gamma^*(k)$. Let

$$\mu := \frac{t\lambda}{a}.$$

The definition of t gives $1 - \mu = t(1-\lambda)/b$, so $\mu \in (0,1)$. By convexity of A_f, $\mu e_g + (1-\mu)e_h \in A_f$. Moreover,

$$\mu e_g + (1-\mu)e_h = tk + (1-t)x^*.$$

Hence $t \in \Gamma^*(k)$.

It remains to show that $t \leq \alpha$ for every $\alpha \in \Gamma^*(k)$. Fix $\alpha \in [0, t)$ and let $\gamma := \alpha/t$, so $\gamma \in [0,1)$. By the minimality of a and b, respectively,

$$\gamma e_g + (1-\gamma)x^* = \gamma a g + (1-\gamma a)x^* \notin A_f$$

and

$$\gamma e_h + (1-\gamma)x^* = \gamma b h + (1-\gamma b)x^* \notin A_f.$$

Using $\mu e_g + (1-\mu)e_h = tk + (1-t)x^*$, we obtain

$$\begin{aligned} \alpha k + (1-\alpha)x^* &= \gamma\big(tk + (1-t)x^*\big) + (1-\gamma)x^* \\ &= \mu\big(\gamma e_g + (1-\gamma)x^*\big) + (1-\mu)\big(\gamma e_h + (1-\gamma)x^*\big). \end{aligned}$$

Since $\mu \in (0,1)$, Claim 2 implies that $\alpha k + (1-\alpha)x^* \notin A_f$. Thus no $\alpha < t$ belongs to $\Gamma^*(k)$. Together with $t \in \Gamma^*(k)$, this shows that $t = \min \Gamma^*(k)$. Therefore,

$$\frac{1}{\min \Gamma^*(k)} = \frac{1}{t} = \frac{\lambda}{a} + \frac{1-\lambda}{b},$$

which is the desired equality. ■

Claim 13 *For every $g, h \in A_f$ and $\lambda \in [0,1]$,*

$$\lambda g + (1-\lambda)h \sim_{A_f} q_{\lambda\alpha_g + (1-\lambda)\alpha_h}.$$

Proof. By Claim 10, there exist $\alpha_g, \alpha_h \in [0,1]$ such that $g \sim_{A_f} q_{\alpha_g}$ and $h \sim_{A_f} q_{\alpha_h}$. Since $g, h \in A_f$, the definition of $\succsim_{A_f}$ implies that $q_{\alpha_g}, q_{\alpha_h} \in A_f$. The characterization in Lemma 1 then gives

$$\min \Gamma^*(g) = \min \Gamma^*(q_{\alpha_g}), \qquad \min \Gamma^*(h) = \min \Gamma^*(q_{\alpha_h}).$$

Applying Claim 12 to g, h and to $q_{\alpha_g}, q_{\alpha_h}$ yields

$$\begin{aligned}
\frac{1}{\min \Gamma^*(\lambda g + (1-\lambda)h)} &= \frac{\lambda}{\min \Gamma^*(g)} + \frac{1-\lambda}{\min \Gamma^*(h)} \\
&= \frac{\lambda}{\min \Gamma^*(q_{\alpha_g})} + \frac{1-\lambda}{\min \Gamma^*(q_{\alpha_h})} \\
&= \frac{1}{\min \Gamma^*(\lambda q_{\alpha_g} + (1-\lambda) q_{\alpha_h})} \\
&= \frac{1}{\min \Gamma^*(q_{\lambda\alpha_g + (1-\lambda)\alpha_h})}.
\end{aligned}$$

Here $\lambda q_{\alpha_g} + (1-\lambda) q_{\alpha_h} = q_{\lambda\alpha_g + (1-\lambda)\alpha_h}$. Hence $\lambda g + (1-\lambda)h$ and $q_{\lambda\alpha_g + (1-\lambda)\alpha_h}$ have the same minimum in their Γ^* sets. Both acts belong to A_f, so their Γ_* sets are also equal. The characterization in Lemma 1 therefore implies

$$\lambda g + (1-\lambda)h \sim_{A_f} q_{\lambda\alpha_g + (1-\lambda)\alpha_h},$$

as required. ■

Claims 9–13 allow us to define $V : \mathcal{F} \to [0,1]$ by $V(g) = \alpha_g$. This function represents $\succsim_{A_f}$ and is affine on A_f. We now establish affinity on all of $\mathcal{F}$.

Claim 14 *For every $g, h \in \mathcal{F}$ and $\lambda \in [0,1]$,*

$$V\big(\lambda g + (1-\lambda)h\big) = \lambda V(g) + (1-\lambda)V(h).$$

Proof. We first establish a preliminary fact. For every $g \in \mathcal{F}$, there exists $\bar{\gamma} > 0$ such that, for every $\gamma \in [0, \bar{\gamma}]$,

$$\gamma g + (1-\gamma)x_* \in A_f \quad \text{and} \quad V\big(\gamma g + (1-\gamma)x_*\big) = \gamma V(g).$$

Moreover, if $g \notin A_f$, then

$$\bar{\gamma} = \gamma_* := \frac{\theta}{\alpha_g},$$

where $V(g) = \alpha_g$. By Claim 11, $V(q_\lambda) = \lambda$ for every $\lambda \in [0,1]$. In particular, $V(x_*) = 0$ and $V(q_\theta) = \theta$.

If $g \in A_f$, take $\bar{\gamma} = 1$. Since $x_* \in A_f$ and A_f is convex, the mixture belongs to A_f for every $\gamma \in [0,1]$. Moreover, V is affine on A_f and $V(x_*) = 0$, so the displayed equality holds.

Now suppose that $g \notin A_f$. Since $g \sim_{A_f} q_{\alpha_g}$, $\Gamma_*(g) = \Gamma_*(q_{\alpha_g})$. Also, $q_{\alpha_g} \notin A_f$; otherwise, $q_{\alpha_g} \succsim_{A_f} g$ would imply $g \in A_f$. Thus $\alpha_g > \theta$, and Claim 11 gives

$$\Gamma_*(q_{\alpha_g}) = \{\beta \in [0,1] : \beta\alpha_g \le \theta\} = \left[0, \frac{\theta}{\alpha_g}\right].$$

Hence $\Gamma_*(g) = [0, \gamma_*]$, where $\gamma_* := \theta/\alpha_g > 0$, and the required mixtures belong to A_f for every $\gamma \in [0, \gamma_*]$.

The act $\gamma_* g + (1-\gamma_*)x_*$ is a boundary point of A_f, since γ_* is the largest weight on g for which this mixture belongs to A_f. The act q_θ is also a boundary point, by Claim 11. Since A_f and its complement are both convex (by No-Hedging and Claim 2), they are separated by a hyperplane. Membership of a mixture with either benchmark depends only on the value of the separating affine functional at the act being mixed. Therefore, any two boundary points are indifferent under $\succsim_{A_f}$, and in particular

$$\gamma_* g + (1-\gamma_*)x_* \sim_{A_f} q_\theta.$$

It follows that

$$V\big(\gamma_* g + (1-\gamma_*)x_*\big) = V(q_\theta) = \theta.$$

For any $\gamma \in [0, \gamma_*]$,

$$\gamma g + (1-\gamma)x_* = \frac{\gamma}{\gamma_*}\big[\gamma_* g + (1-\gamma_*)x_*\big] + \left(1 - \frac{\gamma}{\gamma_*}\right) x_*.$$

Both acts on the right belong to A_f. Hence, by the affinity of V on A_f and $V(x_*) = 0$,

$$\begin{aligned} V\big(\gamma g + (1-\gamma)x_*\big) &= \frac{\gamma}{\gamma_*}\theta + \left(1 - \frac{\gamma}{\gamma_*}\right) V(x_*) \\ &= \frac{\gamma}{\gamma_*}\theta = \gamma \alpha_g = \gamma V(g), \end{aligned}$$

where the third equality follows from $\gamma_* = \theta/\alpha_g$. This proves the preliminary fact.

We now prove that V is affine on all of $\mathcal{F}$. Take arbitrary $g, h \in \mathcal{F}$ and $\lambda \in [0,1]$. Apply the preliminary fact to g, h, and $\lambda g + (1-\lambda)h$. Since each associated upper bound is strictly positive, choose a common $\gamma > 0$ small enough that all three mixtures with x_* belong to A_f and the preliminary equality applies to each of them. Observe that

$$\begin{aligned} &\gamma\big[\lambda g + (1-\lambda)h\big] + (1-\gamma)x_* \\ &\quad = \lambda\big[\gamma g + (1-\gamma)x_*\big] + (1-\lambda)\big[\gamma h + (1-\gamma)x_*\big]. \end{aligned}$$

Since the two acts on the right belong to A_f, the affinity of V on A_f gives

$$\begin{aligned} V\Big(\gamma\big[\lambda g + (1-\lambda)h\big] + (1-\gamma)x_*\Big) &= \lambda V\big(\gamma g + (1-\gamma)x_*\big) \\ &\quad + (1-\lambda)V\big(\gamma h + (1-\gamma)x_*\big) \\ &= \lambda\gamma V(g) + (1-\lambda)\gamma V(h). \end{aligned}$$

Applying the preliminary fact to the left-hand side gives

$$\gamma V\big(\lambda g + (1-\lambda)h\big) = \lambda\gamma V(g) + (1-\lambda)\gamma V(h).$$

Since $\gamma > 0$, division by γ yields

$$V\big(\lambda g + (1-\lambda)h\big) = \lambda V(g) + (1-\lambda)V(h),$$

as required. ■ ■

Lemma 4 *The relation $\succsim_{A_f}$ satisfies monotonicity.*

Proof. We first show that $\succsim_{A_f}$ satisfies Independence. Since V represents $\succsim_{A_f}$ and is affine on $\mathcal{F}$, for every $g, h, j \in \mathcal{F}$ and $\lambda \in (0,1)$,

$$\begin{aligned} V\big(\lambda g + (1-\lambda)j\big) - V\big(\lambda h + (1-\lambda)j\big) &= \lambda V(g) + (1-\lambda)V(j) \\ &\quad - \lambda V(h) - (1-\lambda)V(j) \\ &= \lambda\big(V(g) - V(h)\big). \end{aligned}$$

Since $\lambda > 0$, this implies

$$g \succsim_{A_f} h \quad \Longleftrightarrow \quad \lambda g + (1-\lambda)j \succsim_{A_f} \lambda h + (1-\lambda)j.$$

Thus $\succsim_{A_f}$ satisfies Independence.

We next show that $\succsim_{A_f}$ satisfies one-side monotonicity. We first note that, for every $j \in A_f$, there exists $s \in S$ such that $j(s) \in A_f$, identifying the lottery $j(s)$ with its constant act. Indeed, $f \in c(A_f) = c(A_f \cup \{j\})$. The contrapositive of One-Side Monotonicity therefore gives some $s \in S$ such that $f \in c(A_f \cup \{j(s)\})$. By maximality of A_f, this implies $j(s) \in A_f$.

Now take arbitrary $g, h \in \mathcal{F}$ and suppose that $g(s) \succsim_{A_f} h$ for every $s \in S$. Fix $\alpha \in [0,1]$ and $x \in \{x_*, x^*\}$, and suppose that

$$\alpha g + (1-\alpha)x \in A_f.$$

Applying the preceding observation to this act, there exists $s \in S$ such that $\alpha g(s) + (1-\alpha)x \in A_f$. Since $g(s) \succsim_{A_f} h$, the definition of $\succsim_{A_f}$ implies that $\alpha h + (1-\alpha)x \in A_f$. This holds for every such α and x, so $g \succsim_{A_f} h$. Hence $\succsim_{A_f}$ satisfies one-side monotonicity.

By Riella (2014), Independence and one-side monotonicity imply monotonicity. Therefore, $\succsim_{A_f}$ satisfies monotonicity. ■

For each $f \in \mathcal{F}$, define the collection of maximal menus at which f is chosen by

$$\mathcal{S}_f := \{A \in \mathcal{A} : f \in c(A) \text{ and } \nexists B \in \mathcal{A} \text{ such that } A \subsetneq B \text{ and } f \in c(B)\}.$$

By the maximal-menu result established above, $\mathcal{S}_f \neq \varnothing$ for every $f \in \mathcal{F}$. The set $\mathcal{S}_f$ need not be a singleton: the result guarantees a maximal menu, not a maximum menu. Thus distinct maximal menus $A_f, A'_f \in \mathcal{S}_f$ may be incomparable under set inclusion; that is, neither $A_f \subseteq A'_f$ nor $A'_f \subseteq A_f$ need hold.

For every $f \in \mathcal{F}$ and every $A_f \in \mathcal{S}_f$, let $\succsim_{A_f}$ denote the binary relation constructed above from the menu A_f. Define the collection of all such relations by

$$\mathcal{L} = \left\{\succsim_{A_f} : f \in \mathcal{F},\ A_f \in \mathcal{S}_f\right\}.$$

By the Anscombe–Aumann representation theorem, for every $f \in \mathcal{F}$ and every $A_f \in \mathcal{S}_f$, choose a probability–utility pair $(\pi_{A_f}, \bar{u}_{A_f}) \in \Delta(S) \times \mathbb{R}^X$ representing $\succsim_{A_f}$. Claim 9 then implies that, for every such pair and every $x \in X$,

$$\bar{u}_{A_f}(x^*) \geq \bar{u}_{A_f}(x) \geq \bar{u}_{A_f}(x_*).$$

Moreover, $x^* \succsim_{A_f} x_*$, while $x_* \not\succsim_{A_f} x^*$: the reverse comparison fails at $\alpha = 1$ and benchmark x_* because $x_* \in A_f$ but $x^* \notin A_f$. Hence $\bar{u}_{A_f}(x^*) > \bar{u}_{A_f}(x_*)$. For each f and A_f, define the normalized

utility

$$u_{A_f}(x) := \frac{\bar{u}_{A_f}(x) - \bar{u}_{A_f}(x_*)}{\bar{u}_{A_f}(x^*) - \bar{u}_{A_f}(x_*)}, \qquad x \in X.$$

This positive affine transformation preserves the represented relation. The induced collection of probability–utility pairs is therefore

$$\mathcal{M} := \left\{ (\pi_{A_f}, u_{A_f}) : f \in \mathcal{F},\ A_f \in \mathcal{S}_f \right\}.$$

By construction, every $(\pi_{A_f}, u_{A_f}) \in \mathcal{M}$ satisfies

$$u_{A_f}(x^*) = 1 \geq u_{A_f}(x) \geq u_{A_f}(x_*) = 0 \qquad \text{for every } x \in X.$$